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LoRe: Personalizing LLMs via Low-Rank Reward Modeling

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Personalizing large language models (LLMs) to accommodate diverse user preferences is essential for enhancing alignment and user satisfaction. Traditional reinforcement learning from human feedback (RLHF) approaches often rely on monolithic value representations, limiting their ability to adapt to individual preferences. We introduce a novel framework that leverages *low-rank preference modeling* to efficiently learn and generalize user-specific reward functions. By representing reward functions in a low-dimensional subspace and modeling individual preferences as weighted combinations of shared basis functions, our approach avoids rigid user categorization while enabling scalability and few-shot adaptation. We validate our method on multiple preference datasets, demonstrating superior generalization to unseen users and improved accuracy in preference prediction tasks.

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1 Introduction

Aligning Large Language Models (LLMs) with human values is paramount for enhancing their relatability and effectiveness. Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017) is the standard approach to achieve this alignment. However, conventional approaches often rely on monolithic value representations, which inadequately address the diverse needs of various populations (Bakker et al., 2022; Durmus et al., 2023).

In recent years, there has been a growing advocacy for pluralistic alignment in AI systems. Researchers (Sorensen et al., 2024; Kirk et al., 2024a; Jang et al., 2023) emphasize the importance of designing AI systems that cater to the unique requirements of individuals and groups. This paradigm shift has spurred the development of novel methods, benchmarks, and training datasets. Nevertheless, many existing approaches depend on pre-selected diversity-defining dimensions—such as demographics (Moon et al., 2024; Kwok et al., 2024), personality traits (Castricato et al., 2024; Jiang et al., 2023; Serapio-García et al., 2023; Zhu et al., 2024), and writing styles (Han et al., 2024; Jang et al., 2023; Bai et al., 2022)—which categorize individuals into predefined groups, potentially overlooking intra-group variability. The scarcity of large-scale preference datasets has previously hindered personalized LLM development. However, pioneering efforts by (Kirk et al., 2024b; Zollo et al., 2024) have facilitated the exploration of personalization methods beyond predefined user types.

Early attempts to personalize LLMs involved integrating additional inputs—typically learnable models that generate latent representations of user preferences based on past interactions—into the design of LLMs (Li et al., 2024; Chen et al., 2024b; Woźniak et al., 2024) or reward models (Poddar et al., 2024; Chen et al., 2024a). These strategies, however, often need substantial individual user data or rely on categorizing users based on factors such as demographics, personalities, etc. To address these limitations, we introduce **LoRe**, a novel *Low-Rank Reward Modeling* framework for few-shot personalization.

LoRe leverages a structured low-rank decomposition of reward functions. This approach allows us to model individual preferences as weighted combinations of the basis reward functions, enabling scalable and statistically efficient adaptation with minimal user-specific data. In contrast to prior approaches, LoRe demonstrates superior generalization capabilities to diverse unseen users while maintaining computational efficiency suitable for real-world deployment. By integrating seamlessly with multi-objective alignment frameworks, LoRe

supports personalized response generation without the need for extensive retraining.

2 Preliminaries

A crucial step in aligning LLMs through Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017; Ouyang et al., 2022) is learning a reward function that captures human preferences. Unlike traditional supervised fine-tuning, which relies on explicitly labeled data, RLHF enables models to learn from human comparative judgments. This is particularly valuable in settings where direct supervision is impractical, such as optimizing AI systems for subjective qualities like helpfulness or coherence. In practice, human annotators provide feedback by ranking responses to the same prompt, and this data is used to train a reward model that assigns a numerical score to each response.

A common framework for modeling such preferences is the Bradley and Terry (1952) (BT) model. The BT model represents preferences by assigning scalar scores/rewards to items—where an "item" could be any decision, option, or response. Given two items i and j with scores r_i and r_j, the probability that item i is preferred over item j follows:

$$\mathbb{P}(i \succ j) = \frac{1}{1 + \exp(-(r_i - r_j))}. \tag{1}$$

In the context of LLMs, the reward function maps prompt-response pairs to a scalar score, indicating response quality. This function, typically represented as $r_\phi : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, is trained using human-labeled preference data. Here, ϕ is the reward parameterization in the function class Φ. Specifically, for a given prompt $x \in \mathcal{X}$, if human annotators prefer response $y_c \in \mathcal{Y}$ over $y_r \in \mathcal{Y}$, the BT model expresses the probability of this preference as:

$$\mathbb{P}(y_c \succ y_r | x) = \frac{1}{1 + \exp\left(-(r_\phi(x, y_c) - r_\phi(x, y_r))\right)}. \tag{2}$$

Given a dataset $\mathcal{D}$ of pairwise preference feedback consisting of independent samples, where each sample is a triplet (x, y_c, y_r), and $y_c \in \mathcal{Y}$ is the response preferred over $y_r \in \mathcal{Y}$ for the prompt $x \in \mathcal{X}$ drawn uniformly at random. The joint likelihood of the dataset is:

$$\prod_{(x, y_c, y_r) \in \mathcal{D}} \mathbb{P}(y_c \succ y_r | x). \tag{3}$$

Assuming preferences follow the Bradley Terry Model Eq. (1), the parameters of the reward model can be learned by minimizing the negative log-likelihood defined as:

$$\min_{\phi \in \Phi} \sum_{(x, y_c, y_r) \in \mathcal{D}} \log\left(1 + \exp\left(r_\phi(x, y_r) - r_\phi(x, y_c)\right)\right) = \min_{\phi \in \Phi} \sum_{(x, y_c, y_r) \in \mathcal{D}} \ell(r_\phi(x, y_c) - r_\phi(x, y_r)), \tag{4}$$

where $\ell(z) = \log(1 + \exp(-z))$ is the logistic loss function.

3 Preference Personalization using LoRe

While the BT model assumes a single underlying reward function shared across users, real-world preferences often exhibit significant variation due to individual experiences, biases, and cultural contexts. Next we present an overview of classical work on collaborative ranking, which provides a way to model diverse user preferences.

3.1 Collaborative Ranking from Pairwise Comparisons

Collaborative ranking (Koren et al., 2009) leverages preference data from multiple users to infer individual preferences across a large item set. Each user provides feedback on only a few item pairs, and the goal is to reconstruct their full preference profile by utilizing shared information across users. This approach accounts for diverse preferences, by avoiding the need to aggregate conflicting opinions into a single reward function. Instead, it models personalized rewards by learning structured representations of user preferences.

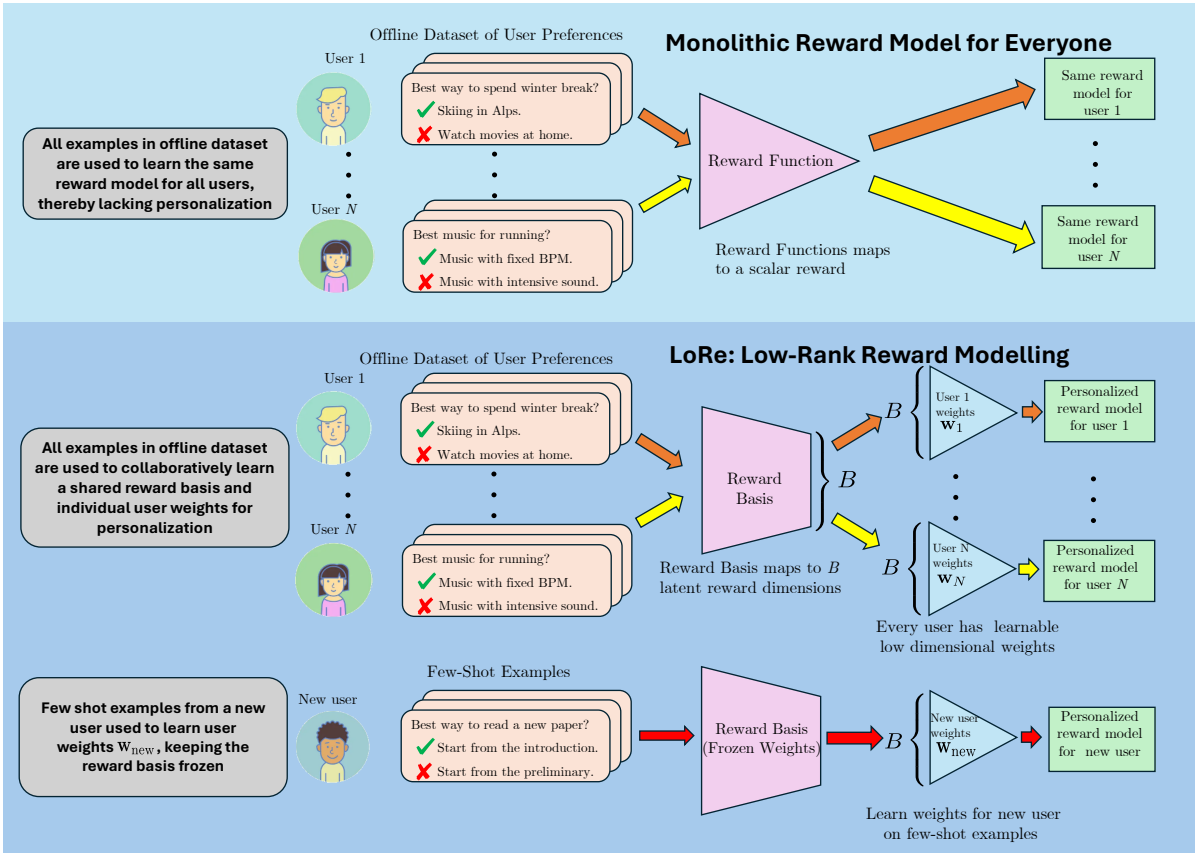

Figure 1 Typically preference data from diverse users is pooled together to train a single reward model for everyone. **LoRe** introduces a more flexible approach by collaboratively learning a shared reward basis from user data. Instead of producing a single reward, this basis generates B latent rewards, which can be combined using a B-dimensional weight vector unique to each user to produce personalized rewards. This allows for seamless personalization with minimal effort. For new users, only the user weights need to be learned from few-shot examples, while keeping the reward basis fixed, enabling an efficient and lightweight personalized reward model.

Consider N users and M items, where preferences are captured in a matrix $\mathbf{P} \in \mathbb{R}^{N \times M}$. The ith row of $\mathbf{P}$, denoted $\mathbf{p}_i^\top$, represents user i's rewards across all items. The probability of user i preferring item c over item r is given by the BT model Eq. (1) based on the reward difference $(\mathbf{p}_{i,c} - \mathbf{p}_{i,r})$. Since users provide comparisons for only a small subset of items, recovering the full matrix $\mathbf{P}$ is challenging. A common solution (Lu and Negahban, 2015; Park et al., 2015)—taking into account the similarity among users and items—assumes $\mathbf{P}$ is **low-rank** (has rank $B \ll \min\{M, N\}$) and can be factorized as:

$$\mathbf{P} = \mathbf{W}\mathbf{R}, \tag{5}$$

where the rows of $\mathbf{R} \in \mathbb{R}^{B \times M}$ represent a *reward basis*. and $\mathbf{W} \in \mathbb{R}^{N \times B}$ contains user-specific weights. Each user's preference vector is then given by:

$$\mathbf{p}_i = \mathbf{w}_i^\top \mathbf{R}. \tag{6}$$

Here, $\mathbf{w}_i^\top$ determines how user i's preferences combine the basis vectors in $\mathbf{R}$. The objective is to learn this low-rank matrix from a small fraction of observed pairwise comparisons, enabling personalized and scalable reward learning.

3.2 Low-Rank Reward Modeling for LLM Alignment

Collaborative ranking, by exploiting the low-rank structure in user preferences, enables few-shot learning on a new user even when only a handful of comparisons are available. The idea is to fix the reward basis $\mathbf{R}$